



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

**DIPLOMA in INFORMATION &
COMMUNICATIONS TECHNOLOGY**

APPLICATIONS DEVELOPMENT THEORY 3

**Assessments Guide
2022**

COURSE CODE:	ADT372S
NQF LEVEL:	7

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1. INTRODUCTION

Welcome to Applications Development Theory 3. I trust that you will enjoy this subject and complete it successfully. During this year this subject will be assessed in a variety of ways including tests, exams, assignments, tutorials and a project. It is important that you always do your own work and participate well in group work. This guide contains details of the key assessments you are required to complete this year.

1.1 General Instructions:

During semester 1 you will have to complete a project based on a Restaurant System. This project will consist of several assignments which will take you through the different stages in the Software Engineering process. Details on the Project brief, due dates and mark sheets are provided below.

During semester 2 you will do 2 assignments. The assignment briefs, rubrics and due dates for this are provided below.

1.2 Assessments Breakdown:

There will be one exam per semester. These exams are summative assessments for that semester. Students who miss an exam may take the semester Exception/Sick test provided you adhere to the following conditions: -

- You notify your lecturer that you have missed the exam as soon as possible. Please consult the student handbook to read up on the details for missing an exam.
- Next, log a request on the CRIMS system to take the Exception/Sick test. This process will require reasons and documentation to accompany your ticket submission.
- Students can only qualify to take one Exception/Sick test per subject per year. For more details on this process and the rules please consult CPUT's Student Handbook.

	ASSESSMENT BREAKDOWN AND WEIGHTS						
	1st SEMESTER			2nd SEMESTER			TOTAL
ASSESSMENT CRITERIA	1st TERM	2nd TERM		3rd TERM	4th TERM		
Nature of assessment	Assignments	Assignments	Test	Assignments	Assignments	Test	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	Yes	Yes	100%
Total weight for the assessment	5%	15%	30%	10%	10%	30%	
Will the assessment be moderated?	No	No	Yes	No	No	Yes	

2. PROJECT INFORMATION

2.1 Project Rules

This is a group project with several deliverables, and it is governed by the following rules:

- ✓ Each group will consist of a maximum of **4** members. These members need to come from your class group.
- ✓ Each group should appoint a group leader. This role however will not be static. Instead, it will rotate within the group so that each student gets a turn to lead the group. To ensure transparency one of the group's first tasks will be to create an equitable schedule which indicates the breakdown and duration of each member's leadership role.
- ✓ Students are expected to establish and manage their groups themselves. This includes the self-enrollment of groups onto Blackboard. In exception cases the lecturer will mediate group concerns.
- ✓ You may choose to do this project on your own but remember the outcomes and expectations for the entire project remain the same as if you worked in a group. Therefore, this option is not recommended.
- ✓ While classes are offered remotely all assignment submissions must be done via Blackboard. Should the university resume with contact teaching full time then some hand-ins and activities will be done in person. Notification of changes will occur when conditions change. Other online formats may also be used for project demonstrations, but this option needs the pre-approval of your class lecturer.
- ✓ All group members must be present and contributing during demonstrations of project work. A demonstration schedule indicating the time slot for group presentations will be made available by the class lecturer.

2.2 Project Brief

You are the software manager of a consultancy and are responsible for the following software project. The owners of a small family-owned restaurant have approached you to develop a system for their changing situation. The first restaurant has become hugely popular and the company was in the process of opening up 2 more restaurants to cope with demand when the COVID-19 pandemic surfaced. Due to resource-constraints the first restaurant operated in a basic, non-automated and paper-based way. While some of the “old fashioned” aspects will remain (such as greeting customers at the door) others, such as the tracking and updating of tables physically on a chalkboard diagram and taking orders with pen and paper needs to go. Similarly, the delivery of paper orders to the kitchen for food preparation by waiters is out of date. Once the food is done, the piece of paper is saved for “proper” record keeping by the management. While this system worked in a single small restaurant with a capacity for 30 diners, it yielded a large amount of receipts, wasted a lot of time and way out-of-date for the 2 new and much larger restaurants planned for, which can accommodate 60 diners at a time.

Initially, this project only aimed to automate many of the restaurant operations so that all information pertaining to customer orders and staff activity will be conveniently shared and stored over the restaurants' intranet. Hosts would be able to view table status with a click of a button and the wait staff would be able to enter the customer's orders quickly and efficiently and then have it delivered to the kitchen electronically. The kitchen staff would be able to view the incoming orders and notify the proper wait staff when the food is ready. Bus boys would be able to view the real-time floor status allowing them to know which tables are clean, dirty, or occupied. All the restaurant information should be organized and saved in the database for the management viewing and archival. Since data will be collected and processed automatically up-to-the-minute analysis will be available on

- revenue and revenue percentages per menu item
- Personnel efficiency
- Average turnaround time (how long customers spend in the restaurant)
- Average preparation time (time from when the order is placed to when it is ready) and management will be able to focus on analyzing the data rather than calculating it.

Now that lockdown restrictions have eased the plan now is to resume this project and a solution is needed which addresses the above needs while being compliant with the COVID-19 requirements for businesses to operate in lockdown level 1 in South Africa.

Domain Fieldwork

You should start this project by familiarizing yourself with the environment that this business operates in and how it functions. This could be done by visits to local restaurant(s), interviewing personnel or other team members to learn from their experiences. The aim is to understand the current practice and identify the opportunities where automation can help improve productivity and reduce errors. This will be useful when describing and justifying your design decisions. In software engineering, personal familiarity with the problem domain is one of the greatest assets, and you should show it explicitly in your reports. You will also need to perform an economic study to find the optimal kind of devices to be used by the restaurant personnel. On the one hand, waiters could have handheld PDAs to avoid re-typing the order. On the other hand, fewer fixed

computers on the restaurant floor can be used by multiple waiters. Or, should it be something else? You also need to consider the possibility of loss and damage, particularly for handheld devices.

2.3 Project Assignments

You are required to produce the following deliverables: -

Assignment 1 – The Jam Session			
Time-boxed Class exercise	20 minutes	Due date: -	BB upload on 21/2/2022
Conceptualizing the Requirements (aka The Jam Session). This will be executed during class and should consist of a single page containing the summarized system requirements determined during the brainstorming session. Your answer should include details of how you will address the problems in the old system and what you will include in the new system. To get started use this table of questions below to guide you brainstorming session. Make your system requirements are PRECISE and DETAILED.			
Questions to kick start the Jam Session Who are the people this the system is meant for? What is the core functionality of the system? When and how would the system be used? What is the best way or place to reach an audience for this system? And why? Why is this system important?			
TASK I: Requirements Engineering			Marks
1. Self-enrollment of groups before 21/2/2022			0 or -1
2. The Jam Session 1 – Conceptualization (The Strategy Meeting)			10
Assignment 2			
The Requirements Engineering Specification. This document should include the requirements as indicated below and the conceptual design. It should be in a PowerPoint format and not exceed 8 slides.			
TASK II: Requirements Engineering			Marks
2.1 RE Specification explaining the purpose, scope, stakeholders, feasibility, project roadmap and ethical policy			10
2.2 Use case diagram			10
Assignment 3			
Design, Prototyping and Testing. Two portfolios are required for this task. The design portfolio should include designs for the data, architecture, component and interface. The testing portfolio includes the development of test cases, a code review, the demonstration of a software component and the project retrospective.			
TASK III: Design, Development, Testing			Marks
1. Data Design (ERD)			10
2. Design a storyboard for your proposed solution indicating the key system functionality			10
3. Interface Design (wireframe) for User Story to Place an Order – including demo of Prototype using MarvelApp			10
4. Systems Sequence diagram for User Story to Update a Table status			10
5. Architectural Design			10

6. Test cases User Story to Calculate the Bill	10	2/5/22
7. Peer test of component for User Story to Calculate the Bill	10	9/5/22
8. Component demonstration	10	tbd
9. Project Retrospective	10	23/5/22

3. ASSIGNMENTS for SEMESTER 2

3.1 Assignment 3 Brief and mark sheet

Assignment 4

This group assignment extends the Restaurant system of the project started in semester 1. It is based on the following assumptions: -

- ✓ You have implemented the system that you worked on in semester 1 as a webApp which has been in operation for a while.
- ✓ The company now wants this system to be a mobileApp which must be up and running by the end of this month. Given this sudden change to both software and business platforms and short deadline discuss your plans for evolving the system based on the following :-

	Question	Marks
1	Explain the change processes that the software would have to undergo for the mobileApp to be implemented at such short notice.	10
2	According to Lehman and Belady software will always change. Based on Lehman's laws of Software evolution explain which 2 laws are the most important for evolving the current system to the new mobileApp. Provide reasons for your selections.	10
3	You have decided that the company needs a sustainable way of managing changes to systems. Create a Legacy management plan which will ensure that aging systems are able to continue to work successfully as the business environment changes.	15
4	Pecha Kucha Presentation	15
	Total	50

This assignment counts 10% toward your marks for ADT372S. Each group must upload a single Pecha Kucha presentation to BB on **20/8/2022**. Marks will be allocated as follows:-

Rubrics for Assignment 4				
Task	10 ... 8	7... 4	3 ... 0	Scored

Implementing the process of change	Complete, concise and easy to read discussion of the processes to be followed in this instance so that the changes can be incorporated into the system smoothly	Partial or incorrect explanation of the process to be done to assist in the changes being implemented into this system	Basic or poorly defined change implementation process		
Task	10 ... 8	7 ... 4	3 ... 0	Scored	
Application of Lehman's laws to the evolution of this system	Correct laws identified and its applicability is motivated completely and concisely	Partially correct laws selected and/or application for evolution is incomplete or incorrect	Basic/incomplete/poorly defined application of the laws and/or laws selected incorrect		
Task	15 ... 10	9 ... 5	4 ... 0	Scored	
Legacy systems management plan	Complete, concise and easy to read explanation of how legacy systems will be managed	Partial or incorrect explanation for managing legacy systems in this business	Basic or incorrect plans for design which present difficulties for use		
Task	15 ... 10	9 ... 5	4 ... 0	Scored	
Pecha Kucha presentation	Presentation includes detailed storytelling via text, audio and images; excellent transitions between slides	Partial or incorrect presentation containing text, audio, images and/or working transitions	Incomplete or poorly designed presentation with limited text, audio, images or transitions		
TOTAL				/50	

3.2 Assignment 4 Brief

Assignment 5 - The context

This is a reflexive assignment focused on examining the use of **Contact tracing apps** during the COVID-19 Pandemic.

The Coronavirus is a highly infectious airborne disease which causes respiratory illness. From the end of 2019 it swept across the world swiftly and left a trail of severely ill people and fast climbing death tolls. The World Health Organization, governments, healthcare and other specialists; along with many others sought

- to inform and prepare citizens for this new unknown health threat and
- they looked for ways to combat its devastating effects and flatten the curve of the spread.

Digital contact tracing was seen as an easy way for tracking the disease using mobile phones to determine the contact between the phone user and an infected patient. During May of this year

Google and Apple jointly released a Contact tracing feature which connected to existing health apps on users' mobile phones as part of a software upgrade. At that stage contact tracing apps had already been in widespread use in many east Asian countries. Later, **the COVID Alert SA App** was launched by the South African government in July to supplement manual contact tracing in the country. In September 2020 as the national lockdown moved to level 1 the president urged all citizens to use this app. It was felt that the careful monitoring that contact tracing apps provided is useful in managing the spread of the Coronavirus and for quicker preventative measures to be put in place.

Assignment Brief - 5a

Based on the above you, as a software developer, are trying to make sense of this phenomenon from a **sociotechnical perspective** as you hope to contribute to similar such solutions. To do so you need to read widely on this app and other similar contact tracing apps to understand their purpose, how it works, the key elements, impacts and potential they offer. A few sources have been provided below to get you started with this process. Thereafter, you will need to discuss Contact tracing systems from a sociotechnical perspective by relating the various aspects of such apps to the different layers in the Sociotechnical Systems Stack (STS) as defined by Sommerville. This assignment has two parts. In this first section your answers need to focus on **the software engineering stack** and the **organization**. Currently, in lockdown level 1 South Africans can enjoy more freedoms in movement and assembly; and businesses and other organizations are changing some of its practices to suite the new compliance regulations and try to recover from financial losses caused by the earlier lockdown levels. However, with these changes come the potential for the virus to spread easily and quickly again and for fatalities to increase to overwhelming proportions. It is against this background that you need to discuss the various elements which are found in each of the sociotechnical layers mentioned below and explain what they are, how they fit in and how it contributes to the bigger picture of monitoring and preventing the spread of the Coronavirus.

Assignment Brief - 5b

In this assignment the reflexive exercise continues with the examination of **Contact tracing apps** used during the COVID-19 Pandemic. The focus of this assignment is centred on the ethical and legal aspects of contact tracing apps. You will also need to round off this assignment by presenting conclusions to the use of contact tracing apps generally, for COVID 19 use and in South Africa.

3.3 Assignment 4 rules

Assignment rules

This is a group assignment for which there are 2 deliverables. Here are the rules –

- **Groups.** Each group should consist of **no more than 4 members**. The group members should remain constant for both assignment deliverables. Each group should submit one entry to BB. Assignment 4a is due by **27/9/2022** and the submission for assignment 4b is due by **14/10/2022**.
- **Weight.** Assignment 4 counts 10% of your remaining assignment marks for ADT372S.
- **Your solutions.** Your solution should be presented in the form of a technical report. This is a formal document which should be well formatted, easy to read and all sources used should be correctly referenced. This report should not exceed 4 pages so use the bulk of it for content. A sample guide for a technical report is included in the websites listed below.
- Here are a few **websites** to help you start your information gathering process: -

<https://www.bbc.com/news/technology-52807635>

<https://covid19.apple.com/contacttracing>

<https://www.google.com/covid19/exposurenotifications/>

<https://www.businessinsider.co.za/covid-alert-sa-app-south-africas-official-coronavirus-app-2020-9>

<https://medium.com/technical-writing-is-easy/how-to-write-technical-report-e935210002c9>

3.4 Assignment 4 Rubrics

Rubric 5a				
Task	5 ... 4	3 ... 2	1 ... 0	Score
Report format and introduction	Grammatically correct with good layout and space is used optimally. Clear, concise introduction	Report contains language errors and/or a flawed layout and/or use of space, introduction	Report contains language errors, layout and use of space is poor and/or introduction lacking	
Task	10 ... 8	7 ... 4	3 ... 0	Score
Discussion of the Equipment required	Numerous paragraphs explaining the key equipment needed by contact tracing apps	Several paragraphs describing the equipment needed by contact tracing apps	Limited/skimpy explanation of the equipment needed by contact tracing apps	
Discussion of the Operating systems needed	Numerous paragraphs explaining the OS requirements for contact tracing apps	Several paragraphs describing the OS requirements for contact tracing apps	Limited/skimpy explanation of the OS requirements for contact tracing apps	
Discussion of the comms and data management	Numerous paragraphs explaining the comms and data management requirements for contact tracing apps	Several paragraphs describing the comms and data management requirements for contact tracing apps	Limited/skimpy explanation of the comms and data management requirements for contact tracing apps	
Discussion of the application system requirements	Numerous paragraphs explaining applications involved with contact tracing apps	Several paragraphs describing the applications involved with contact tracing apps	Limited/skimpy explanation of the applications involved with contact tracing apps	
Discussion of the business processes	Numerous paragraphs explaining the functionality provided by contact tracing apps	Several paragraphs describing the functionality provided by contact tracing apps	Limited/skimpy explanation of functionality provided by contact tracing apps	

Discussion of the organizational considerations	Numerous paragraphs explaining the higher-level strategic activities which affect the operation of contact tracing apps	Several paragraphs describing the higher-level strategic activities which affect the operation of contact tracing apps	Limited/skimpy explanation of the higher-level strategic activities which affect the operation of contact tracing apps	
TOTAL				/65

Rubric 5b				
Task	5 ... 4	3 ... 2	1 ... 0	Score
Report format and references	Good report format. Various electronic and other sources cited correctly.	Report format flawed. Contains some citation errors. Electronic and other sources used.	Poor report format. Citation problems. Only internet sources used.	
Task	20 ... 15	14 ... 9	8 ... 0	Score
Discussion of the societal considerations	Complete, concise and easy to read. Ethical and legal considerations for using contact tracing apps and its future implications given.	Partial or incorrect explanation of the ethical and legal considerations raised on using contact tracing apps for COVID 19 and its implications for the future.	Basic or poor mention of the ethical and legal considerations raised when using contact tracing apps for COVID 19 and its implications.	
Task	10 ... 8	7 ... 5	4 ... 0	Score
Conclusions	Numerous detailed conclusions provided for using contact tracing apps in general and in South Africa.	Several detailed conclusions are reached on using contact tracing apps in general and in South Africa.	A weak conclusion is made on using contact tracing apps general and in South Africa.	
TOTAL				/35